

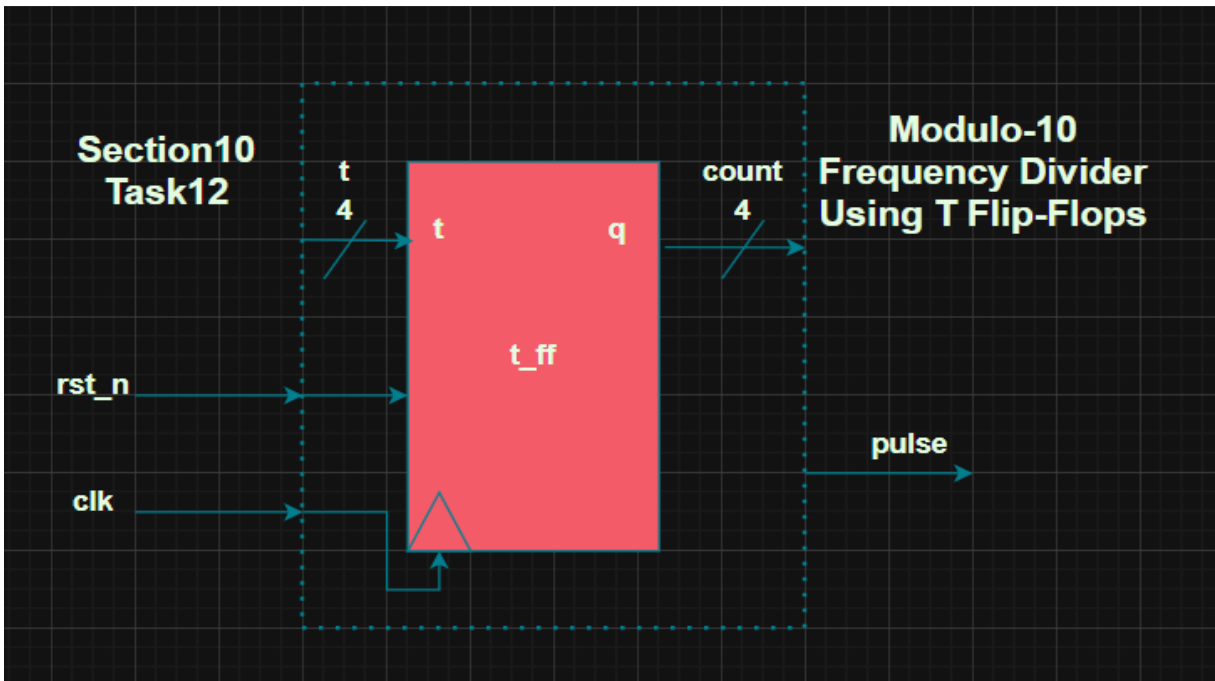
NAME: MUHAMMAD USMAN

REG. NO# 2024-EE-171

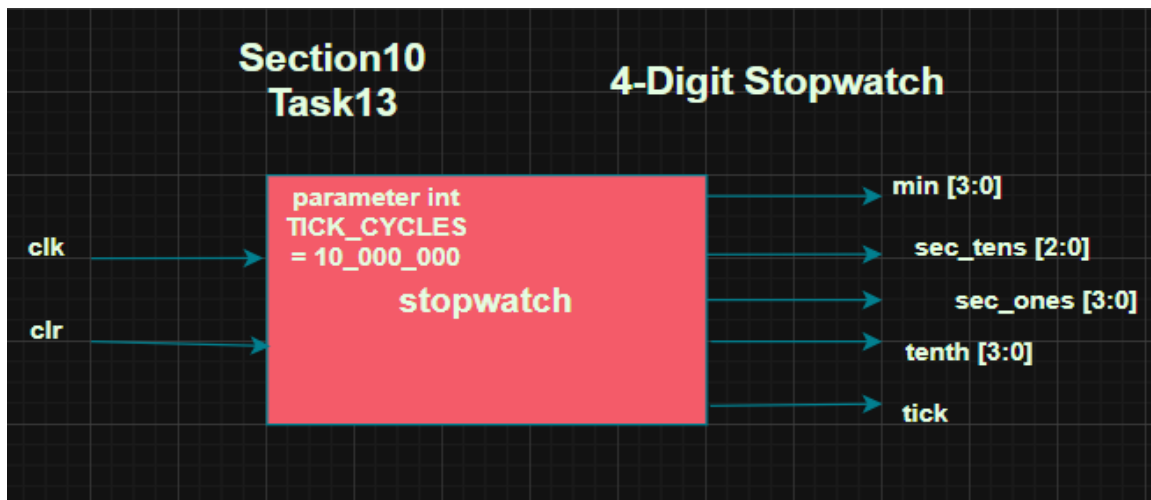
1-EDA Playground link of Task12: https://edaplayground.com/x/Y_su

EDA Playground link of Task13: <https://edaplayground.com/x/KDQx>

2-Schematic Diagram:



Task12_schematic_diagram.png



Task13_schematic_diagram.png

3-Output Screenshot:

```
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164;  Jul 24 06:26 2026
[PASS] Period: 10 cycles
[PASS] Period: 10 cycles
[PASS] Period: 10 cycles
[PASS] Period: 10 cycles
Tests Completed.
$stop at time 505 Scope: mod10_divider_tb.unnamed$$_0 File: testbench.sv Line: 49
ucli% Finding VCD file...
./dump.vcd
[2026-07-24 10:26:15 UTC] Opening EPWave...
Done
```

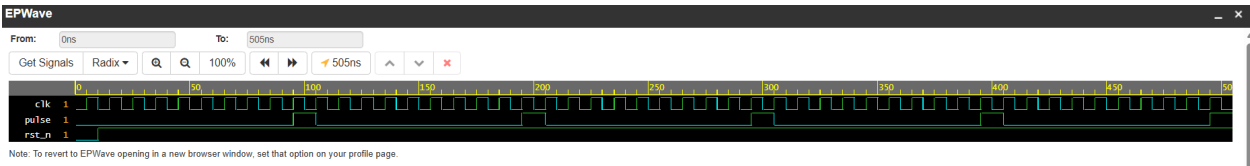
Task12_log_result.png

Log Share

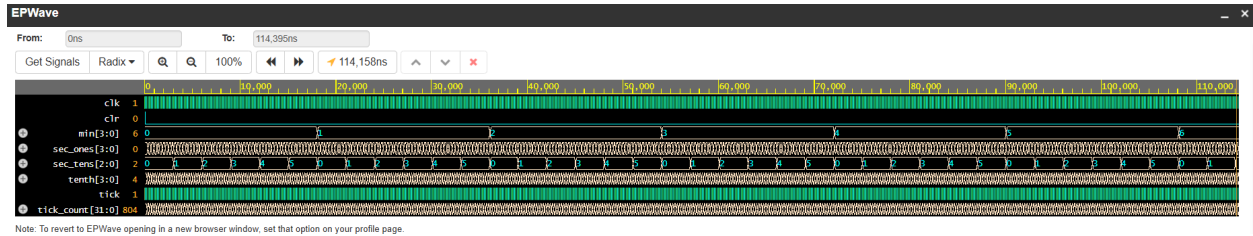
```
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164;  Jul 24 09:27 2026
These Tests are for TICK_CYCLES=3 to make simulation faster
Stopwatch started
[PASS] At tick 10: 0:01:0
[PASS] At tick 60: 0:06:0
[PASS] At tick 600: 1:00:0
[PASS] At tick 6000: 0:00:0
All tests completed.
$stop at time 180036 Scope: stopwatch_tb File: testbench.sv Line: 74
ucli% Finding VCD file...
./dump.vcd
[2026-07-24 13:28:01 UTC] Opening EPWave...
Done
```

Task13_log_result.png

4-Waveform Screenshot:



Task12_waveform.png



Task13_waveform.png

5-Explanation :

Task12 (Calculation for the 0.1 s Tick Generation):

Given:

- Clock frequency $f_{clk} = 100 \text{ MHz}$
- Clock period $T_{clk} = \frac{1}{100 \times 10^6} = 10 \text{ ns}$
- Required tick period $T_{tick} = 0.1 \text{ s} = 100,000,000 \text{ ns}$

Step 1: Number of Clock Cycles per Tick

$$N = \frac{T_{tick}}{T_{clk}} = \frac{100,000,000 \text{ ns}}{10 \text{ ns}} = 10,000,000$$

Therefore, the internal counter must count from **0 to 9,999,999** (10 million unique states) before wrapping and producing a tick.

Step 2: Tick Frequency

$$f_{tick} = \frac{1}{T_{tick}} = \frac{1}{0.1 \text{ s}} = 10 \text{ Hz}$$

Step 3: Tick Pulse Width

- The tick output is a strobe pulse that is high for exactly **one clock cycle**.
- Pulse width = $T_{clk} = 10 \text{ ns}$.

Step 4: Duty Cycle

$$\text{Duty Cycle} = \frac{\text{Pulse Width}}{\text{Tick Period}} = \frac{10 \text{ ns}}{0.1 \text{ s}} = \frac{10 \times 10^{-9}}{1 \times 10^{-1}} = 1 \times 10^{-7}$$